

JEE MAIN REVISION MASTER: HYDROCARBONS

2026 1 QUESTIONS 1.67% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 15 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Hydrocarbons - Core Application	15 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Hydrocarbons - Core Application (Very High Yield — 15 PYQs)

Core Principle: Key problem pattern in Kcet Hydrocarbons.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Statement I: Staggered conformation of ethane is more stable than the eclipsed conformation. Statement II: The torsional strain in staggered..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "The major product formed when 1- Bromo -3- Chlorocyclobutane reacts with metallic sodium in dry ether is..." [Ans: N/A]
- [2024 Shift] Q#1 (MCQ): "2-methyl propane can be prepared by Wurtz reaction. The haloalkanes taken along with metallic sodium and dry ether are :..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Statement I: Staggered conformation of ethane is more stable than the eclipsed conformation. Statement II: The torsional strain in staggered conformation is more. Read the above statements and choose the correct answer from the options given below.

- (A) Both Statement I and Statement II are false
- (B) Both Statement I and Statement II are true
- (C) Statement I is true but Statement II is false
- (D) Statement I is false but Statement II is true

Correct Answer: None

Solution Analysis: Statement I is true. In ethane, the staggered conformation is more stable because the hydrogen atoms on the two carbon atoms are as far apart as possible. The dihedral angle between C-H bonds is 60° , so the repulsion between bonding electron pairs is minimum. Statement II is false. Torsional strain is not more in the staggered conformation. It is maximum in the eclipsed conformation because the C-H bonds of the two carbon atoms are directly aligned with each other. This causes greater repulsion between bonding electron pairs. So, $\text{Statement I is true but Statement II is false.}$